

Telling the time

The earliest people who tried to keep track of time used simple devices such as sundials to track the movement of the Sun through the sky. It was many centuries later, about 700 years ago, that mechanical clocks were introduced in Europe. Faster communications in the 19th century led to timekeeping becoming standardized across the world.

Hour hand indicates the hour of the day

Gilded sun marks the position of the Sun in the Zodiac



Originally, the phases of the Moon were indicated.

Early timekeeping

Sundials

Sundials measure time by using a shadow cast by an upright rod (gnomon) to track the position of the Sun in the sky.



Sundial (9th century) in Northern Ireland

Water clocks

In water clocks, a jar is filled with water, which drains away at an even rate to indicate how much time has passed.

Hourglasses

A specific quantity of sand flows from one glass bulb into another through a narrow opening. The sand flows through at a fixed rate, usually taking an hour.

When the sand has run through to the bottom, the hourglass can be turned upside down.



Hourglasses from the 17th century

Incense clocks

In China and Japan, incense sticks that burn at an even rate were used to measure time.

Candle clocks

Slow-burning wax candles had evenly spaced lines marked on them, which were numbered to mark the hours.



The clock tower and detail of clock face (above) in St. Mark's Square, Venice.

Astronomical clocks

Early mechanical clocks in Europe used a falling weight on a chain and a pendulum to turn a series of gears (see box opposite). They often combined 24-hour clock faces with information about the Sun, Moon, and stars. Their elaborate designs often included moving figures (automata) that danced or rang bells at certain times of the day.

Key events

1500 BCE

Sundials were first used in Egypt and Mesopotamia (now Iraq) thousands of years ago. They are only able to tell the time when the Sun is shining.

1088

In China, Su Song built a water-driven mechanical clock to track the cycles of the stars and planets.



Su Song's water clock

1656

Dutch inventor Christiaan Huygens built the first pendulum clock. Driven by the regular sweeps of a weighted pendulum, it was accurate to within a few seconds a day.

1759

English clockmaker John Harrison built the marine chronometer—a spring-driven timepiece that was accurate over a long time, allowing sailors at sea to calculate longitude (see p.93).